

Design of Si-IGBT Gate Driver for Inverter Applications

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Abstract—Gate driver plays an important role in control and protection of power semiconductor devices in converter structures. A careful design of the gate driver is essential for reliable converter operation. Gate resistor selection and isolation between the power and control circuit are of primary importance. Gate drivers are also expected to offer protection like Active Miller clamp (AMC), soft turn-off, short-circuit detection and under-voltage lockout. In this paper, a comprehensive design and selection of components for the various control and protection aspects of a 10kVA IGBT inverter gate driver is presented. The gate driver is built around ACPL-333J optocoupler IC. The protection schemes are discussed along with the inverter structure for better insight and design. A novel method of gate driver selection based on $\frac{di}{dt}$ provided in the device datasheet is proposed to balance between switching speed and reduced EMI. The proposed method results in a simple analytical expression for gate resistor selection. With the selected gate resistor the device $\frac{di}{dt}$ is obtained same as the datasheet value and is validated experimentally. Important aspects related to selection of an on-board integrated isolated power supply and printed circuit board (PCB) design considerations are also presented. Hardware results are provided for the discussed protection schemes.

Index Terms—IGBT protection, inverter, gate driver.

I. INTRODUCTION

Power semiconductor switches are the essential building blocks of power processing circuits. Inverter, for example, utilizes IGBTs to achieve a controlled sinusoidal ac output from a constant dc voltage source. Inverters find use in applications like motor drives, uninterruptible power supplies (UPS) and utility grid applications [1–3].

Proper control and protection of IGBT is essential for reliable operation of the inverter. For example, IGBTs in an inverter can experience high current flow during load or dc-bus short circuit. Load short-circuit can arise from insulation failure. Dc-bus short-circuit may happen when both top and bottom device from the same leg are turned-on simultaneously, either due to faulty control or due to spurious turn-on. IGBT should be protected from such faults. Short circuit currents of even short duration in the order of a few tens of μs can destroy the device [4]. While it is essential to turn off the device quickly under faults, it is equally required that the $\frac{di}{dt}$ associated with the turn-off process is not too high to cause a higher voltage to appear across the turned off device due

to the stray inductance present in the circuit [5]. To ensure this, a proper control is required. Usually, a soft turn-off is employed that balances between the quick response and a low $\frac{di}{dt}$. These and the other control and protection measures are usually handled by the gate driver besides its primary function of providing isolation between the power device and the control circuitry.

Usual control and protection features of the gate driver includes short circuit detection, gate under voltage detection, avoiding spurious turn-on, soft turn-off on short circuit, ensuring mute time after fault detection and providing fault feedback to the controller.

Several integrated chip-based (IC) gate driver optocouplers are commercially available with the above protection features. The use of an IC makes the gate driver circuitry compact, cost-effective and less prone to switching effects of the power circuit [6]. Though the optocoupler IC has various functionalities in-built, a fully functional gate driver involves several design aspects. The design aspects arise because of the application specific customization required for the gate driver. Specifically, selection of the gate resistor, the design or selection of the isolated power supply, protection settings through external components and PCB layout considerations are of importance.

Gate resistor is selected based on allowable device switching speed while the isolated power supply's rating depends on converter dc-bus voltage and gate power consumption. The gate resistor value decides the IGBT switching speed and current sinking/sourcing capability of the optocoupler. A large gate resistor could lead to higher switching losses in the IGBT. Whereas, too low a gate resistor may result in sourcing/sinking of higher currents from/into the optocoupler. The power supply should be capable of supporting the IC power requirements. A poor choice of gate resistor and an isolated power supply can lead to optocoupler malfunction. The protection schemes are well-drafted in the datasheet. However, an understanding of the conditions under which the protection scheme becomes active in a converter structure is essential. Usually, the gate driver IC's features are to be understood with respect to the converter structure for effective utilization of the protection features available in the IC. PCB layout is another critical aspect of a proper gate driver design. Noise source identification, trace

loop lengths, and adequate creepage and clearance are some of the influential elements to consider for the PCB layout.

This paper discusses the design and development of a gate driver PCB for an example application of a 10 kVA inverter. Gate resistor, isolated power supply and protections are the aspects considered. An analytical approach is presented for the calculation of the gate resistance for a given converter rating. To ensure a compact design, an integrated solution-based isolated power supply is suggested. Selection of the isolated power supply based on the rating of the converter and the optocoupler is discussed. Design related to the various protections that ensure reliability of the IGBT switch for inverter applications are developed. The paper also discusses the PCB layout considerations that can help in ensuring a noise-free and compact design.

II. PROPOSED METHOD OF GATE RESISTOR SELECTION

In IGBTs, the turn-on and turn-off $\frac{di}{dt}$ and $\frac{dv}{dt}$ depend on the device input capacitance (C_{iss}) and the gate resistor (R_g). The turn-on $\frac{di}{dt}$ depends on the time constant $\tau = R_g C_{ge}$, C_{ge} being the gate-emitter capacitance. Large values of R_g lead to slower switching speed and increased losses whereas a smaller value could result in high $\frac{di}{dt}$ and $\frac{dv}{dt}$, leading to EMI issues. **The gate resistor selection should ensure a balance between fast switching and reduced EMI.** The proposed method is to determine the gate resistor that results in the same $\frac{di}{dt}$ as obtained in the device datasheet.

The gate-emitter voltage charging is dependent on C_{ge} and R_g values during the device current rise time. The collector current transition of the device during this duration is governed by the gate-emitter voltage values as shown in Fig. 1. C_{ge} is fixed for a chosen device. C_{ge} value can be obtained from the typical capacitance value versus collector-emitter voltage characteristics [7]. Hence it is possible to relate the current rise time with the gate resistor value alone for a given IGBT and fixed V_{gg} . From Fig. 1, the analytical rise time ($t_{r,a}$) of the device can be derived as $t_{r,a} = t_2 - t_1$. In terms of the device parameters,

$$t_{r,a} = R_g C_{ge} \ln \left(\frac{V_{gg} - V_{ge,th}}{V_{gg} - V_{ge,M}} \right). \quad (1)$$

where, $V_{ge,th}$ is the threshold gate-emitter voltage and $V_{ge,M}$ is the gate-emitter voltage corresponding to rated load current (I_o). The value of C_{ge} , $V_{ge,th}$ and $V_{ge,M}$ are available in the device datasheet.

Therefore,

$$\frac{di}{dt} = \frac{I_o}{t_{r,a}}. \quad (2)$$

With the new load current (I_o), the IGBT would achieve the same $\frac{di}{dt}$ as in the datasheet when

$$t_{r,a} = \frac{I_o}{I_c} t_r \quad (3)$$

where I_c and t_r are the datasheet test current and rise time respectively. Equating (1) and (3) yields the required R_g .

An example of R_g selection for the three-phase 10kVA IGBT based-inverter gate driver is considered. The maximum peak current expected through the IGBT is around 22 A considering a 10 % tolerance at the inverter output nominal voltage (415 V). The selected R_g must ensure that the device maximum di/dt with $I_o = 22$ A is same as mentioned in datasheet. In the IKW15N120T2 datasheet [7] t_r is shown as 25 ns for $I_c = 15$ A and $T_j = 25^\circ\text{C}$. Using this data and (3) we calculate

$$t_{r,a} = \left(\frac{22}{15} \right) 25 \text{ ns} = 36.67 \text{ ns}. \quad (4)$$

For device current of 22 A, with $V_{ge,th} = 7$ V, $V_{ge,M} = 10.5$ V and (4), the required value of gate resistance is calculated as 42Ω using (1). To account for reduction in the resistance due to its tolerance, a slightly higher value of 44.2Ω is used in the hardware. Note that $V_{gg} = V_{out} = 13$ V (considering the 2 V optocoupler internal drop) in (1) for the calculation of the required R_g .

III. SELECTION OF ISOLATED POWER SUPPLY

A power supply is required to handle the gate power requirements and to meet the isolation needs discussed later in Section. IV-E. The required isolation is related to the dc bus voltage. Further, the power supply should ensure a low coupling capacitance between the output and input stages in order to reduce the common-mode noise generated during high dv/dt operation of the power device [8]. Integrated off-the-shelf isolated power supplies having the mentioned features are available and is used in the present work. The required power rating of the power supply is estimated as

$$P_{ps} = (P_g + P_q) \times 1.2 \quad (5)$$

where

$$P_g = Q_{ge} V_{out} f_{sw} \text{ and } P_q = V_{cc2} I_q.$$

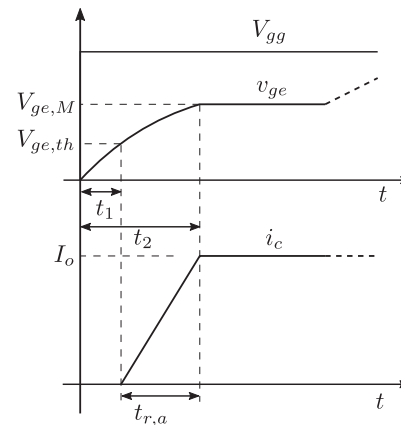


Fig. 1. Turn-on transient of IGBT for the rated load current I_o .

In (5), P_g represents the power required by the device gate while P_q accounts for the power consumed by the gate driver to bias all internal circuit. A safety factor of 20% is considered for the calculation. V_{cc2} and V_{out} are the output voltages of the power supply and the gate driver respectively. $V_{out} \approx V_{cc2} = 15$ V. Q_{ge} is the amount of charge required by the device gate to attain $V_{ge} = 15$ V, f_{sw} is the device switching frequency (10 kHz) and I_q is quiescent current drawn by the gate driver to meet the P_q requirement. For ACPL-333J the maximum I_q is 5 mA [9]. The value of Q_{ge} is approximately 100 nC for the chosen device [7]. Using these data in (5), the worst case power rating of the source P_{ps} is calculated as 180 mW.

MEV1S2415SC, a 1 W 24/15 V power supply with a galvanic isolation of 3 kVDC from Murata Power solutions is chosen as it meets all the requirements.

IV. CONTROL AND PROTECTION USING ACPL-333J

ACPL-333J is a 2.5 A_{peak} output current optocoupler ideally suited for driving IGBTs with ratings up to 1200 V and 150 A [9]. It offers an insulation voltage of 1414 V_{peak} . The IC has a 8.3 mm creepage and clearance between the high and low side pins, avoiding additional cutouts in the PCB to meet the creepage requirement. The following discussion presents a detailed overview of the control and protection features of the optocoupler along with the selection of the external components.

A. Desaturation Detection

A short circuit fault results in a high collector-emitter voltage, v_{ce} under device on state. For a fixed gate-emitter voltage, this takes the device out of saturation (desaturates). The DESAT protection feature uses this voltage information to detect the short-circuit fault. The driver continuously senses v_{ce} and turns off the device when it exceeds a prefixed threshold value (V_{DESAT}) under on state. Being a voltage detection technique, the DESAT feature is also effective in avoiding device failure due to insufficient gate-emitter voltage. Fig. 2(a) shows the external components – a blanking capacitor C_b , DESAT diode D and series resistor R – required for DESAT operation.

It is required that the DESAT protection does not trigger a fault when the device is off or during the turn-on transition. Under these conditions, v_{ce} will be high even without a fault. An internal MOS shorts C_b while the device is off and ensures that the DESAT feature remains inactive for the turn-off duration of the device. The diode D blocks the high v_{ce} voltage and ensures that the DESAT pin of the driver is not connected to the high voltage directly. When the device transitions to the on-state from off-state, C_b is charged slowly using an internal constant current source (I_b) of 240 μ A. The time taken by the capacitor to reach the threshold voltage (V_{DESAT}) of 6.5 V is known as the blanking time (t_{blank}). t_{blank} is set higher than the device voltage fall time and ensures that the fault signal is not triggered during device turn-on transient. **Towards the end of the turn-on transient, diode D starts conducting and the DESAT pin starts measuring v_{ce} directly.** A 100 pF

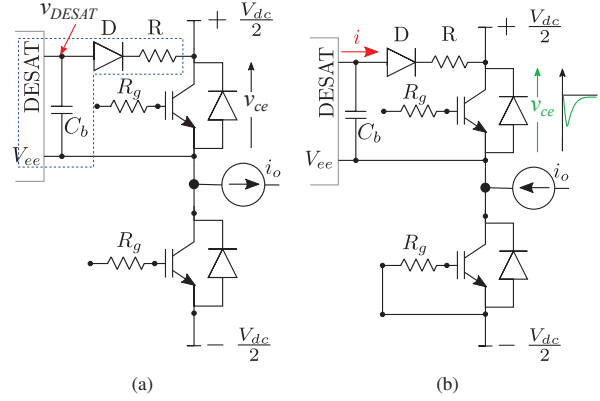


Fig. 2. (a) External components for DESAT operation. (b) Typical operating condition under which series resistor R helps to suppress sourcing of large current from DESAT pin.

capacitor is recommended by ACPL-333J datasheet to achieve $t_{blank} = 2.7 \mu$ s calculated using the relation

$$t_{blank} = \frac{V_{DESAT} \times C_b}{I_b}. \quad (6)$$

When the DESAT feature is active, noise in v_{ce} can lead to voltages higher than V_{DESAT} at the DESAT pin if the DESAT diode junction capacitance is high [10]. For this reason ACPL-333J recommends a diode with lower junction capacitance. Further, a low reverse recovery time t_{rr} for the DESAT diode ensures that C_b is not charged to the threshold during the transition from turn-on to turn-off state of the device. During this transition, the reverse recovery current of DESAT diode can charge C_b for a short duration before the internal MOSFET shorts it out. It is recommended that $t_{rr} < 75$ ns. The resistor (R) is used to avoid sourcing of large current from the DESAT pin due to the voltage overshoot of the IGBT's anti-parallel diode during its turn-on transient. The circuit condition is shown in Fig. 2(b) wherein the current transitions from the bottom IGBT to the top anti-parallel diode.

The worst case fault signal generation time of t_{blank} occurs when an IGBT in a leg turns-on into an existing short-circuit. If the short circuit fault happens when the device is already on, the time taken will be much lower due to the fast charging of the blanking capacitor with the reverse recovery current of the DESAT diode and I_b .

B. Under-Voltage Lockout (UVLO)

IGBTs typically require gate voltages of 15 V to operate in the saturation region. For a fixed i_c , insufficient gate voltage results in a higher on-state drop (v_{ce}) across the device. Also, the device may enter the active region for $v_{ge} < 10$ V, resulting in a much higher on-state drop. For the selected IGBT IKW15N120T2, the effect of insufficient gate voltage on v_{ce} can be observed from its output characteristics for a representative current of 22 A as shown in Fig. 3. Large on-state drop at rated load conditions can lead to device failure due to excess heat generation.

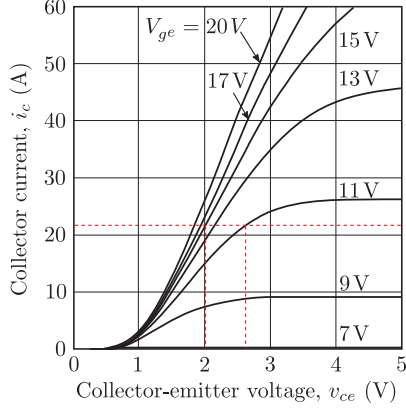


Fig. 3. Typical output characteristics of IGBT IKW15N120T2 at $T_j = 25^\circ\text{C}$ [7].

Fig. 4(a) and Fig. 4(b) show the circuit representation of the UVLO feature and its operating hysteresis band. If V_s falls below UVLO-, V_{out} is shorted to device emitter (V_{ee}) through a low resistance internal strong pull-down device (SPD).

C. Active Miller Clamp (AMC)

IGBT parasitic capacitances can result in unintentional turn-on of the device [11]. Fig. 5 explores one such condition in inverters. Consider the leg to be initially under dead time (Δt) where both the IGBTs are off. The load current (i_o) is conducted by the freewheeling diode of the top device and the dc bus voltage V_{dc} appears across $\bar{S}_a$. Then, when S_a is turned on, the current transfers from the top diode to the bottom IGBT and the voltage across S_a rises from zero to V_{dc} at the rate of few V/ns. This results in a high $\frac{dv}{dt}$ across the Miller capacitor (C_{gc}) of S_a and generates a significant charging current

$$i_M = C_{gc} \frac{dv}{dt}. \quad (7)$$

With the V_{clamp} path absent, the charging current flows through the parallel combination of gate resistor R_g and gate capacitance C_{ge} of S_a . The resultant voltage can cause

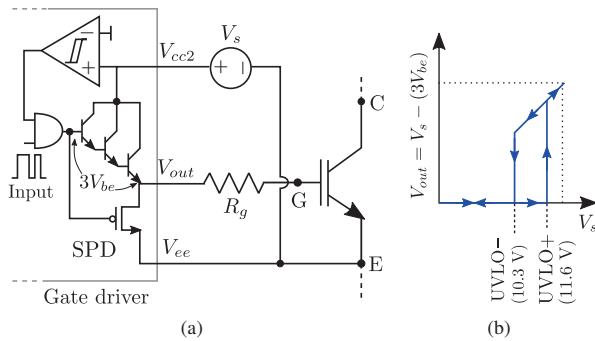


Fig. 4. (a) Equivalent circuit representation of UVLO operation. (b) Characteristic curve demonstrating UVLO behaviour.

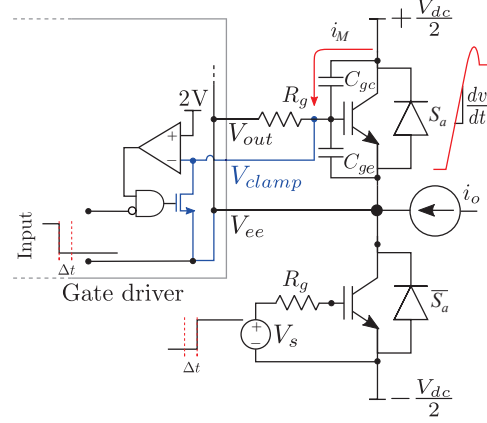


Fig. 5. Effect of Miller capacitor and AMC protection circuitry.

unintentional turn-on of the device if v_{ge} exceeds $V_{ge,th}$. Such spurious turn-on results in a dc-bus short circuit. To avert such spurious turn-on, the AMC feature monitors v_{ge} when the input gate pulse is low and clamps the device gate terminal to its emitter V_{ee} through V_{clamp} pin when v_{ge} falls below a prefixed value. The clamp path diverts i_M from flowing into the parallel combination of R_g and C_{ge} . The clamp is released as soon as the input to the gate driver goes high.

D. Soft turn-off

The R_g value selected in Section. II assumes normal operation. The R_g along with the gate-emitter capacitance C_{ge} sets the time gate charging and discharging time duration irrespective of the device current. During turn-off under a short circuit fault, the current being large, the same R_g will result in high $\frac{di}{dt}$. The high $\frac{di}{dt}$ causes a large voltage drop (v_{L_s}) across the loop stray inductance L_s as shown in Fig. 6.

$$v_{ce} = V_{dc} - v_{L_s} = V_{dc} - L_s \frac{di}{dt} \quad (8)$$

As the turn-off process reduces the current, the $\frac{di}{dt}$ is negative and hence the stray inductance drop adds up with the dc bus voltage and appears as large momentary voltage overshoot across the device being turned-off. Simultaneous large voltages and currents appearing even for short duration can lead to device failure. ACPL-333J uses a soft turn-off feature to handle this.

Fig. 6 shows the circuit representation of the soft turn-off feature. As shown, a strong pull-down device (SPD) and a weak pull-down device (WPD) are used respectively during normal and short circuit fault conditions. The higher resistance of WPD helps in achieving a slower turn-off during fault. From the ACPL-333J datasheet, the resistances of SPD (R_{SPD}) and WPD (R_{WPD}) are,

$$R_{SPD} = \frac{V_O}{I_{OL}} = \frac{2.5}{1.5} = 1.67 \Omega. \quad (9)$$

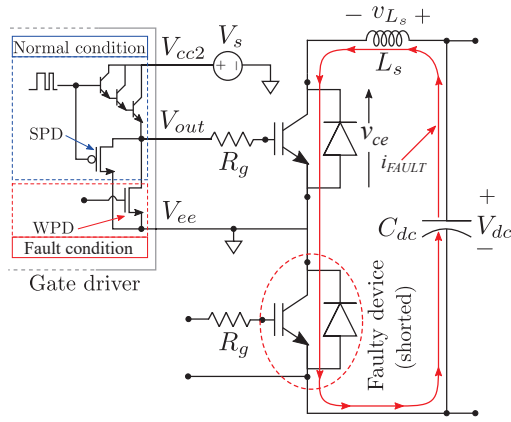


Fig. 6. Equivalent circuit representation of soft turn-off feature of ACPL-333J.

$$R_{WPD} = \frac{V_O}{I_{OLF}} = \frac{14}{0.14} = 100 \, \Omega \quad (10)$$

E. Isolation

To understand the need for isolation between the DSP and the power devices, Fig. 7(a) considers one leg of a three-phase four wire inverter with dc bus mid point connected to the grid neutral (N). The DSP is also referenced to the same neutral N. This leads to short-circuit of the bottom dc-bus capacitor (path ①), loss of control over bottom IGBT ($\bar{S}_a$) due to shorting of its collector and emitter terminals (path ②) and dc-bus short-circuit when the top IGBT (S_a) is turned on (path ③). Therefore, isolation becomes necessary between S_a and DSP, $\bar{S}_a$ and DSP and between the emitters of S_a and $\bar{S}_a$ as shown in Fig. 7(b). ACPL-333 J offers a maximum working working insulation voltage of $1414 V_{peak}$.

F. Fault feedback

ACPL-333J has a fault feedback ($\overline{FAULT}$) that can be fed to the DSP. A fault will send a low signal to the DSP. The internal auto reset of ACPL-333J brings back the $\overline{FAULT}$

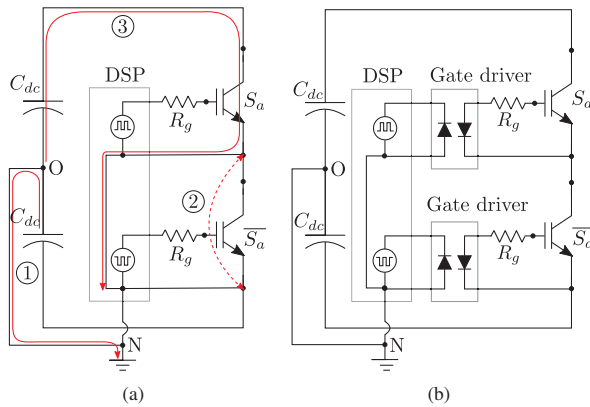


Fig. 7. (a) Non-isolated gate driver. (b) Isolated gate driver.

to level high after a mute time. The auto reset feature of ACPL-333J saves some additional hardware implementation that would have been required for restarting the converter operation after fault removal.

V. PCB LAYOUT MEASURES

The presence of stray inductance in the gate-emitter loop can result in ringing in the gate-emitter voltage. The total stray inductance in a loop configuration can be calculated by [12] as

$$L_s = \frac{\mu_o}{\pi} \left(l \ln \frac{2d-w}{w} + d \ln \frac{2l-w}{w} \right). \quad (11)$$

From (11) it is clear that the gate-emitter loop configuration (l and d) must be as small as possible. In the developed gate driver PCB, the gate-emitter loop stray inductance (L_s) is estimated to be around 16 nH using (11). The damping factor (ζ) of the gate-emitter loop is calculated to be 5.25 for $R_g = 44.2 \, \Omega$, $C_{ge} = 1 \, \text{nF}$ and $L_s = 16 \, \text{nH}$. The high damping factor ensures good damping of gate-emitter voltage.

A minimum creepage of 8 mm should be maintained between the signal side and power side ground planes and the pins connected to them, to retain the effective isolation of the IC. The spacing also helps in reducing parasitic coupling between the power and signal side grounds. The large ground planes also provides a low-impedance path that reduces the induced noise voltages due to the flow of common-mode (CM) currents generated by the switching action.

The parasitic coupling capacitor between the input and output stages of the isolated power supply generates large CM currents due to high dv/dt action [8]. Therefore, the input side ground of the power supply is kept separated from the control signal ground plane to avoid interference with the control signal. Also, length of the sensitive traces are kept minimum to lower EMI. The clearance and creepage is followed as per IPC-2221B standard.

VI. HARDWARE RESULTS

The developed gate driver was tested using a double-pulse test setup for a device current of 22 A and dc-bus voltage of 800 V. Fig. 8 shows the device turn-on transient when the gate resistor is set to $44.2 \, \Omega$ as calculated in Section. II. From Fig. 8 it is seen that the device rise time is 40 ns and is very close to the desired value of 36.67 ns. Fig. 9 and Fig. 10 show the protection scheme operation of ACPL-333J. From Fig. 9, the role of DESAT in fault detection and device protection can be observed. The time taken for fault isolation is $4.4 \, \mu\text{s}$ and is higher than the $t_{blank} = 2.7 \, \mu\text{s}$. The longer time is due to the turn-off delay of internal MOS which is parallel to the DESAT capacitor and I_b tolerance [13]. From Fig. 10, it is evident that the AMC feature of ACPL-333J keeps v_{ge} within permissible limit to avoid spurious turn-on of the top IGBT. The typical gate-emitter threshold value for IKW15N120T2 is 5.8 V [7]. The PCB layout of the developed gate driver is depicted in Fig. 11. It can be seen that the gate loop configuration is very small. Also, a compact design is obtained with the dimensions being $51.3 \, \text{mm} \times 60.32 \, \text{mm}$.

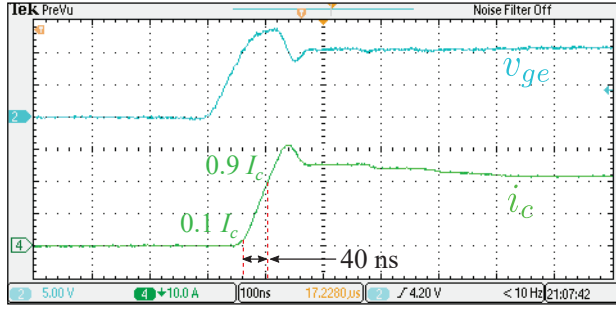


Fig. 8. Turn-on transient of gate-emitter voltage (v_{ge}) and collector current (i_c) for $I_c = 22$ A and $R_g = 44.2 \Omega$.

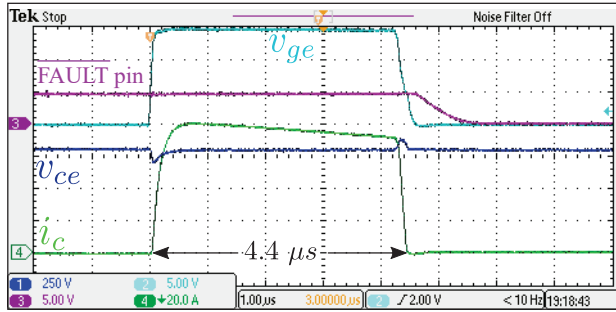


Fig. 9. Waveform showing short-circuit fault and action of DESAT protection.

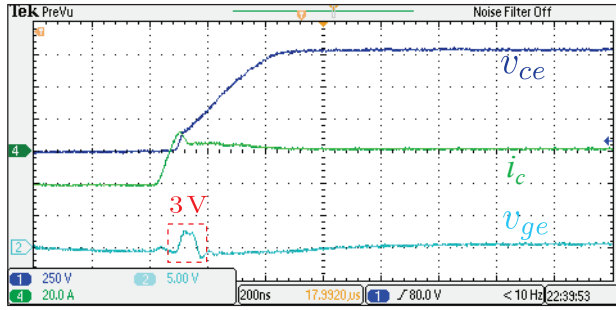


Fig. 10. Waveform showing AMC action of top IGBT gate driver under the condition described in Section IV-C.

